

Project Proposal: E-voting system

1.Introduction

The E-voting system is a web based website which is designed to conduct political elections digitally in a secure, efficient, and transparent manner in Nepal. It allows the registered voters of the country to cast their votes while ensuring the confidentiality. The system replaces traditional physical voting systems with a modern solution which saves time, reduces the cost and minimizes errors.

2.Problem Statement

Currently, Voters face a lot of problems such as:

- 1.Walking far distances to vote
- 2.Hectic and tedious voting systems

Which leads to many citizens not being able to vote, or experiencing a hectic and tedious voting experience.

This project aims to create an online approach for voters through which a citizen can vote just a few clicks away.

3.Objectives:

- To develop a secure online voting platform
- Ensure a transparent and fair election
- Save time for both the voters and the staff involved
- Provide a real time vote count with accuracy
- To prevent duplication and illegal voting.

4. Literature Review

4.1 Introduction

Electronic Voting(E-Voting) systems are digital platforms that allow elections to be conducted online instead of traditional paper-based methods. These systems aim to improve efficiency, transparency, accuracy and accessibility in the elections.

4.2 Review of Existing Systems and Theories

Several countries have explored electronic voting systems to improve election efficiency and accessibility.

In 2005, Estonia became the first nation to implement nationwide internet voting. Estonian election officials declared the electronic voting system a success and found that it withstood the test of real-world use.

Research on e-voting systems shows that they can improve:

- Election efficiency
- Faster result generation
- Reduction of manual errors
- Better accessibility for voters

However, studies also identify challenges such as:

- Cybersecurity risks
- Infrastructure requirements
- Public trust in online systems

In the context of Nepal, geographical barriers and limited digital infrastructure highlights the need for a simple and practical e-voting solution.

Therefore, this project focuses on developing a basic academic model of a web-based E-voting system emphasizing usability, transparency, and efficiency

4.Methodology:

Requirement Identification:

The studying and analysis of existing voting systems(electionbuddy, simply voting) along with their limitations. Identification of user needs such as a secure and confidential login system, vote casting, management of the candidates and proper and accurate result generations.

Feasibility Study:

Evaluating the practicality of the system from 3 different perspectives:

Technical Feasibility: checking if required technology is suitable.

Operational Feasibility: Ensuring that the system is user friendly and a easy UI.

Economic Feasibility: Analyzing the cost effectiveness of this system compared to the traditional voting system.

High Level Design:

Designing the system architecture and workflow:

- Define the user module, admin module and database.
- Creating the basic flow (Login to vote to store to count to display results)
- Plan the database structure (the voters, candidates, the number of votes)
- Outline the interaction between frontend and backend

Development:

- Frontend development and design using HTML, CSS and Javascript.

- Backend development using php
- Database using MySQL

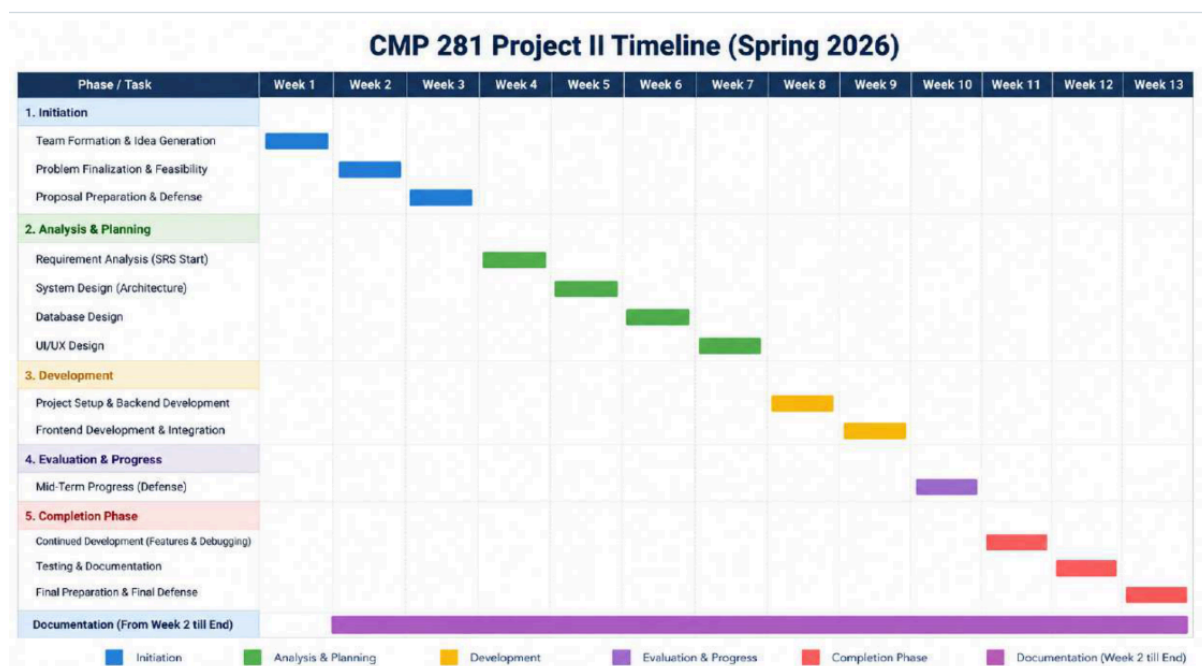
Testing:

- Testing of all modules
- Testing security and authentication
- Use testing for user's experience and interface

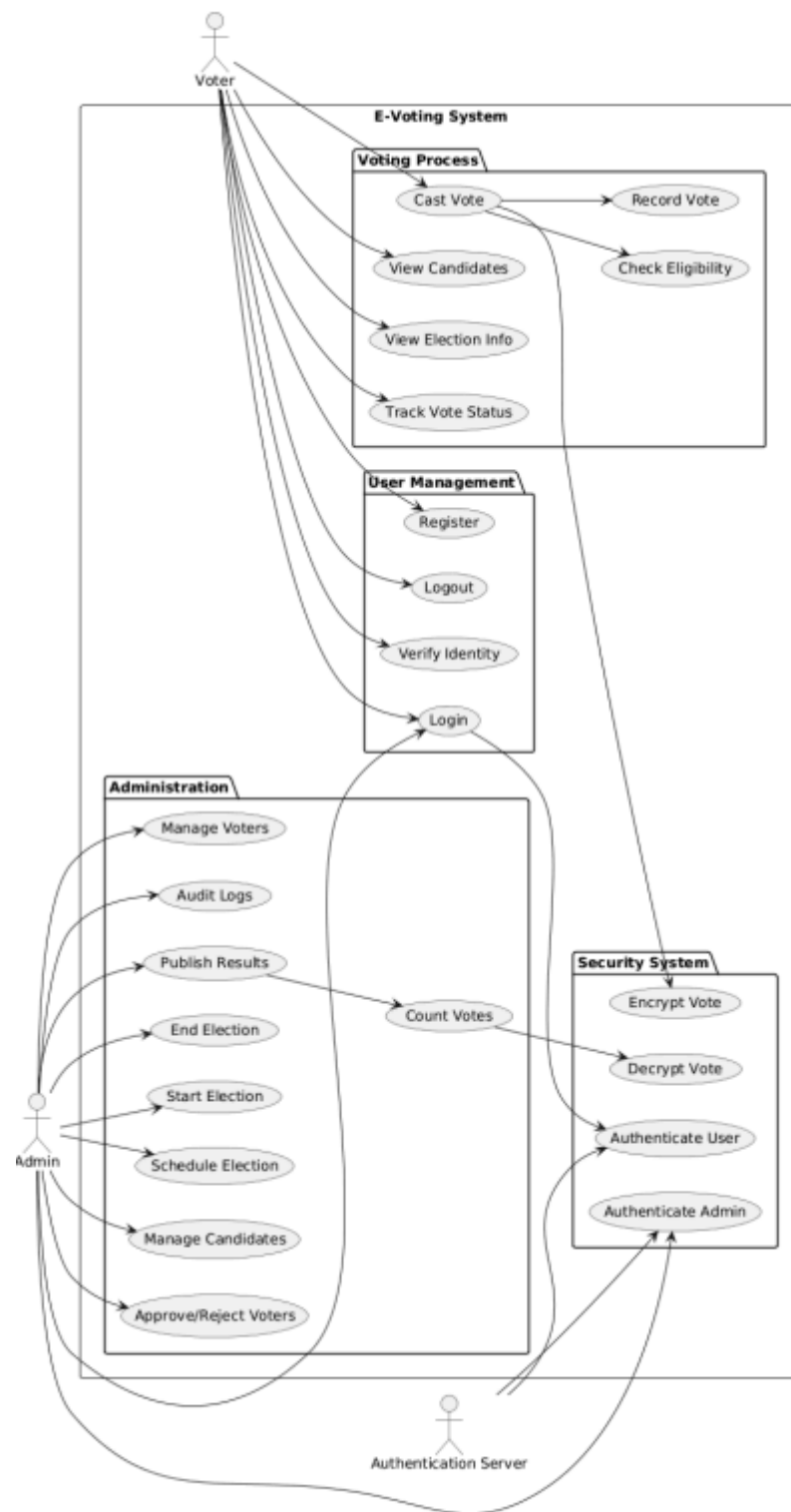
Deployment:

Deploy the system on a local server and make it accessible and usable to users.

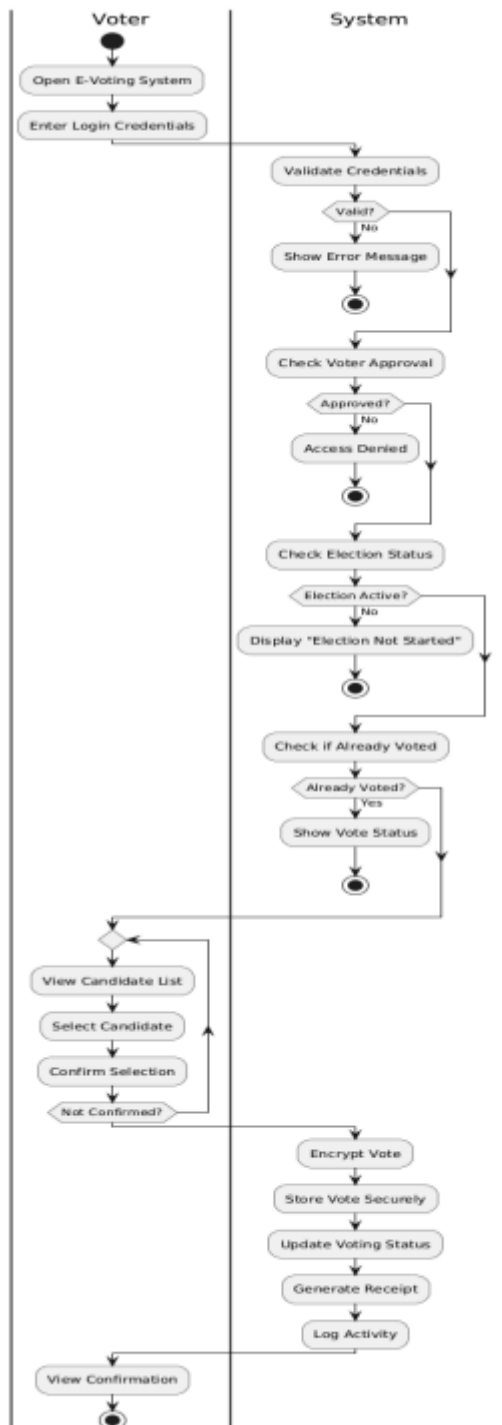
5.Gantt Chart:



6. Use Case Diagram of E-Voting System in Nepal



7. Flowchart of the EVoting System



6. Expected Outcome:

- A secure and reliable online voting system
- Accurate and real time vote count
- Prevention of fraudulent voting (ensuring one vote per voter)
- Improving cost effectiveness and efficiency to traditional systems.

7. References

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